

Peiqi Gao

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EDUCATION

Washington University in St. Louis

Ph.D. in Computer Science, advised by Chenyang Lu

St. Louis, MO

Aug. 2024 – Present

Washington University in St. Louis

M.S. in Computer Science

St. Louis, MO

Aug. 2023 – Present

Wuhan University

B.Eng. in Computer Science and Technology, Hongyi Honor College, GPA 3.63/4.00

Hubei, China

Aug. 2019 – Jun. 2023

RESEARCH INTEREST

Machine Learning | Computer Vision | Vision-Language Models | Reinforcement Learning

PUBLICATIONS

Wang, R., Wang, Z., **Gao, P.**, Li, M., Jeong, J., Xu, Y., Lee, Y., Baum, C. M., Connor, L. T., & Lu, C. (2025).

Real-Time Video-based Human Action Recognition on Embedded Platforms. ACM Transactions on Embedded Computing Systems (TECS), 24(5), Article 135.

Wang, R., **Gao, P.**, Lynch, P. J., Liu, T., Lee, Y., Baum, C., Connor, L. T., & Lu, C. (2025). *CHEF-VL: Detecting Cognitive Sequencing Errors in Cooking with Vision-language Models*. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT). (Accepted)

PROJECTS

Smart Kitchen | Machine Learning, Vision Language Model, Embedded System

Sep. 2023 – Present

- Developed an online video-based human action recognition (HAR) system designed to assist users with cognitive impairments to complete tasks independently.
- Finetuned **vision-language models** (Qwen2.5VL, LLaVA-video, etc) on our dataset to improve the performance of HAR.
- Implemented reinforcement learning with a vision language model to improve HAR performance.

AI-Driven Diagnostics for MDS | Machine Learning, Object Detection

Sep. 2024 – Present

- Developing an end-to-end deep learning framework to automate the diagnosis of MDS from pathology slides.
- Implemented a quality control pipeline using **DenseNet** to preprocess gigapixel Whole Slide Images (WSIs) and select high-quality cell patches for downstream analysis.
- Integrated **YOLO** for cell detection and **U-Net** for precise instance segmentation to isolate single cells from complex backgrounds.
- Designed a **Vision-language model-based** method for analyzing segmented cells and predicting disease probability.

TECHNICAL SKILLS

Languages: Python, C/C++, Java

Frameworks & Tools: PyTorch, Transformers, LLaMA-Factory, OpenCV, Git, Docker

Research Areas: Human Action Recognition (HAR), Vision-Language Models, Object Detection, Medical Imaging